PRACTICAL PROBLEMS

PROBLEM 1 (practical)

You are required to investigate seven inorganic compounds.

Your test-tube rack is numbered 1 to 9. Two of the positions are empty. Each of the seven test-tubes provided contains only one compound in aqueous solution. Using only these solutions, pH indicator paper, and test-tubes, you are to identify as many of the ions present as you are able.

For your information, record in the table the observations you make on mixing the solutions. Use the following symbols:

elimination reactions: $\downarrow$ precipitate; $\uparrow$ gaseous product;

$\downarrow_s$ precipitate soluble in the excess of a precipitating agent.

colours: w - white or colourless, b - blue, g - green, y - yellow, p - pink, r - red, br - brown.

pH: a - acidic, b - alkaline, n - neutral.

Equipment:

A home-made rack contained 9 test-tubes with the unknown solutions, 30 empty Wassermann-tubes and one small beaker containing the pH indicator paper. Into each solution a dropper was inserted, and thus, the test-tubes need not to be removed from the rack while handling them. According to the original plan the following nine unknown solutions were expected to be given to the participants: $CoCl_2$, $Fe(SCN)_3$, NH_4OH, KI, $AgNO_3$, Na_2HAsO_4, $HgCl_2$, $NiCl_2$, $CuCl_2$.

During the discussion of the International Jury it became known that in some countries the corresponding laws forbid the pupils in secondary schools to handle mercury and arsenic compounds. For this reason these two compounds were removed from the rack and consequently the number of ions to be detected - and the marks available - were reduced to 12 (from the original 15). (Under these conditions the alkali and nitrate ions cannot be detected.)

The order of the test-tubes varied individually, but the first two contained invariably red solutions ($CoCl_2$ and $Fe(SCN)_3$), while the last two were the green $NiCl_2$ and $CuCl_2$ symbolizing the Hungarian national colours, red-white-green.

SOLUTION

The ions of the remaining seven solutions can easily be identified by mutual reactions. Out of the 21 possible reactions, 12 are common positive reactions. Additional information is available from the colour of 4, and the smell of one solution.

$AgNO_3$: reacts with all the six compounds;

NH_3: with the exception of iodide it gives a characteristic reaction with all the others salts;

$Fe(SCN)_3$: its colour and reaction with NH_3, I^-, Ag^+ are characteristic;

$CoCl_2$: can be detected from its colour and by adding NH_3 or Ag^+;

KI: can be identified by its reaction with Ag^+ and from the evolution of I_2 due to an addition of Fe^{3+} or Cu^{2+};

$CuCl_2$: can be detected from its colour and reaction with NH_3, I^- and Ag^+;

$NiCl_2$: has a characteristic colour and reacts with NH_3 and Ag^+.

PROBLEM 2 (practical)

You are required to estimate the heat (enthalpy) change on mixing a series of 5 liquids to produce equimolar mixtures and to explain the temperature changes.

Procedure:

1. A mixture of trichloromethane (chloroform) and propanone (acetone)

Measure 0.5 mol of trichloromethane in the measuring cylinder labelled A_1 and measure its temperature. Dry the thermometer with a piece of tissue paper. Measure 0.5 mol of propanone using measuring cylinder B_1, pour it into a beaker and measure its temperature. Record the average of the two temperatures as temperature t_1 (to 0.1 ^oC). Leave the thermometer in the beaker. Add the trichloromethane to the propanon, stir the mixture carefully and follow the temperature changes until a maximum or minimum is reached. Record this extreme temperature as temperature t_2. Dispose the mixture into a special bottle labelled "waste solution", dry the reaction beaker and the thermometer and proceed to the next part of the experiment.

2. A mixture of methanol and propanone (acetone)

Measure 0.5 mol of propanone in measuring cylinder B_1 and 0.5 mol of methanol in measuring cylinder A_2, and continue as in part 1.

3. A mixture of methanol and n-hexane

Measure 0.5 mol of methanol into measuring cylinder A_2 and 0.5 mol of hexane into measuring cylinder B_2, and continue as in part 1.

4. A mixture of methanol and water

Measure 0.5 mol of methanol into measuring cylinder A_2, measure its temperature and pour it into the beaker. Rinse the cylinder thoroughly with distilled water and then measure 0.5 mole of water using this measuring cylinder. Continue as instructed in the above part 1.

Tasks:

Calculate the enthalpy (heat) changes involved in the mixings on the basis of the temperature changes observed. In your calculations you should neglect heat exchanges with the surroundings as well as the heat capacity of the beaker and thermometer. Briefly explain your results in terms of the molecular interactions in the pure liquids and in the mixture, preferably using sketches.

Data:

Substance	Relative molecular mass	Density ($g\ cm^{-3}$)	Molar heat capacity ($J\ K^{-1}\ mol^{-1}$)
methanol	32.04	0.79	80.61
chloroform	119.38	1.49	114.94
acetone	58.08	0.79	124.96
n-hexane	86.18	0.66	190.10
water	18.02	1.00	75.35

PROBLEM 3 (practical)

You are required to determine the concentrations of hydrochloric acid and potassium iodate in the diluted solution containing both.

Procedure:

A solution containing potassium iodate and hydrochloric acid has already been measured into the volumetric flask provided. Fill the flask to the mark with distilled water using the wash bottle, close it with a stopper and shake it thoroughly. Fill the burette with the standard sodium thiosulphate solution using one of the beakers provided. (The exact concentration of the thiosulphate is given on the label of the bottle.)

a) First titration

Pipette a 10.00 cm^3 aliquot (portion) of the solution from the volumetric flask into a glass stoppered conical flask. Dilute it with 10 cm^3 of distilled water, add 1 g (a small spatula end-full) of potassium iodide and acidify with 10 cm^3 of 10 % sulphuric acid using a measuring cylinder. Titrate immediately the iodine formed with the standard sodium thiosulphate solution until the solution in the flask is pale yellow. Add with a pipette 1 cm^3 of starch indicator solution and continue the titration to completion. Repeat the titration twice more and record your readings on the result sheet.

b) Second titration

Pipette a 10.00 cm^3 aliquot of the solution into another glass stoppered conical flask, dilute with 10 cm^3 of distilled water, add 1 g of solid potassium iodide, and leave to stand for 10 minutes. Then titrate the iodine formed using the standard sodium thiosulphate solution, adding 1 cm^3 of starch indicator solution when the mixture is pale yellow. Repeat the titration twice more, recording your readings on the result sheet.

Task:

Calculate the concentration of the HCl and the KIO_3 in the solution that you prepared by dilution (in mol dm^{-3}).

SOLUTION

The reaction:

$$IO_3^- + 5\ I^- + 6\ H^+ = 3\ I_2 + 3\ H_2O$$

occurs to be quantitative both with respect to IO_3^- and H^+. Consequently the first titration (in the presence of sulphuric acid) is suitable for the determination of iodate, while the second one for the determination of the hydrochloric acid content.